

1. DATASET FIELDS REFERENCED IN THIS GUIDE

Field names below match the NWIS drilling_timeseries schema exactly, so this guide can be cross-referenced directly against logged data.

Field Name	Unit	Typical Range	Description
Bit_Type	text	Tricone / PDC (3-7 blade)	Bit design category and blade/cutter configuration in use
Formation	text	Formation_W/X/Y/Z...	Named geological formation being penetrated at the current depth
ROP	m/hr	5 - 30 m/hr	Rate of penetration
Torque	kft-lb	8 - 22 kft-lb	Rotational resistance measured at surface

2. HOW THESE FIELDS RELATE TO EACH OTHER

Bit_Type vs Formation Suitability: Tricone bits are generally favored in interbedded or highly variable lithology and surface/conductor hole; PDC bits are favored in more consistent, drillable shale/sand sequences but are more sensitive to hard, abrasive stringers, where a mismatched Bit_Type can trigger whirl-driven Torque spikes.

Bit_Type vs ROP for the Same Formation: Switching Bit_Type (e.g. 5-blade to 6-blade PDC) for the same Formation typically changes achievable ROP for a given WOB/RPM combination, since blade count and cutter aggressiveness change the mechanical specific energy relationship.

Formation Transitions vs Torque/ROP Step-Changes: A logged Formation change is often accompanied by a step-change in ROP and/or Torque at similar WOB/RPM, since the two fields together let a step-change be attributed to lithology rather than misdiagnosed as a mechanical fault.

Bit_Type Dull Grading vs Formation History: Repeated abrasive wear or whirl-related dull grading for a given Bit_Type in a specific Formation across multiple wells is the basis for recommending a different Bit_Type for that formation on future offset wells.

3. KEY FORMULAS / RULES OF THUMB

Formula / Rule	Notes
Mechanical Specific Energy depends on Bit_Type (via bit area/aggressiveness) as well as WOB, RPM, Torque, ROP	Comparing MSE across different Bit_Type runs in the same Formation is a fair way to evaluate which bit is more efficient.
Expected ROP range should be re-baselined whenever Bit_Type OR Formation changes	Comparing ROP against the wrong baseline (old Bit_Type or old Formation) produces false anomaly alerts.

4. RISK INDICATORS & RECOMMENDED ACTIONS

Observed Pattern (Fields Involved)	Risk Indicated	Recommended Action
Torque rises sharply immediately after entering a new Formation with an unchanged Bit_Type	Bit_Type may be poorly matched to the new Formation's hardness/abrasiveness	Reduce WOB/RPM through the interval; flag for a Bit_Type change on the next run or offset well

ROP consistently below offset-well average for the same Bit_Type and Formation combination	Possible bit wear, incorrect WOB/RPM, or a hole-condition issue	Compare dull grading history for this Bit_Type/Formation pair; consider adjusting parameters before assuming formation is simply harder
Repeated whirl-related dull grading for a specific Bit_Type across multiple wells in the same Formation	Systematic Bit_Type/Formation mismatch	Recommend an alternative Bit_Type (e.g. anti-whirl design) for that Formation in future well planning

5. SAMPLE LOG ROWS (drilling_timeseries SCHEMA)

Well_ID	Depth_MD	Formation	Bit_Type	ROP	WOB	Torque
W005	2738	Formation_W2 (lower)	PDC 6-blade + RSS	9	13	19.6
W004	2600	Formation_X (upper)	PDC 5-blade	8	12	9.5

6. FIELD_ALPHA CROSS-REFERENCE

- W005 (Alpha-05): a PDC 6-blade bit run through Formation_W2's hard calcareous stringers showed above-average wear (I=4, O=3) with whirl-consistent cutter damage, supporting a future recommendation for an anti-whirl Bit_Type in this specific Formation.
- W001 (Alpha-01): an 8-1/2 in PDC 6-blade bit drilling Formation_X through the mud-loss interval showed only minor wear (I=1, O=1), indicating the Bit_Type/Formation pairing itself was not a contributing factor in that event.

7. OPERATIONAL BEST PRACTICES

- Always log Bit_Type alongside Formation so ROP/Torque anomalies can be correctly attributed to lithology, bit wear, or a genuine mismatch.
- Re-baseline expected ROP/Torque ranges whenever either Bit_Type or Formation changes in the log.
- Build a Bit_Type-vs-Formation performance history across offset wells to guide bit selection on future wells rather than relying on a single well's data.

This Drilling Parameter Reference Guide is a synthetically generated reference document produced for the NWIS prototype, aligned to the drilling_timeseries dataset schema. Not associated with an actual drilling operation.